Routledge
Taylor & Francis Group

Innovations in Education and Teaching International

ISSN: 1470-3297 (Print) 1470-3300 (Online) Journal homepage: www.tandfonline.com/journals/riie20

Leveraging learner innovative behaviour through innovation education: The mediating role of entrepreneurial alertness and the moderating effect of university support

Faisal Iddris, Inusah Salifu & Emmanuel Mensah Kparl

To cite this article: Faisal Iddris, Inusah Salifu & Emmanuel Mensah Kparl (2026) Leveraging learner innovative behaviour through innovation education: The mediating role of entrepreneurial alertness and the moderating effect of university support, Innovations in Education and Teaching International, 63:2, 615-632, DOI: 10.1080/14703297.2024.2383831

To link to this article: https://doi.org/10.1080/14703297.2024.2383831

INNOVATIONS IN EDUCATION AND TEACHING INTERNATIONAL
2026, VOL. 63, NO. 2, 615–632
https://doi.org/10.1080/14703297.2024.2383831

Leveraging learner innovative behaviour through innovation education: The mediating role of entrepreneurial alertness and the moderating effect of university support

Faisal Iddris[a], Inusah Salifu[b] and Emmanuel Mensah Kparl[a]

[a]Department of Management Studies Education, Akenten Appiah-Menka University of Skills Training and Entrepreneurial Development, Kumasi, Ghana; [b]Department of Adult Education and Human Resource Studies, School of Continuing and Distance Education, College of Education, University of Ghana, Accra, Ghana

ABSTRACT

While Innovation Education holds significant potential for driving economic development, its specific impact on Learner Innovative Behavior within Ghanaian higher education remains unexplored due to a lack of research in this area. We conducted a cross-sectional survey to explain the relationship between innovation education and learner innovative behaviour, and the role of entrepreneurial alertness and university support. We used a purposive sampling technique to select a sample of 574 participants. Findings revealed a positive relationship between innovation education and learner innovative behaviour, with entrepreneurial alertness and university support playing critical roles in fostering this relationship. The findings imply that solely relying on innovation education may not sufficiently impact learner innovative behaviour. The study, therefore, recommends integrating entrepreneurial alertness and university support to enhance innovation-oriented learning through innovation-focused curricula. By investing in innovation education and supporting student innovators, universities can contribute to the growth of local economies and address broader societal challenges.

KEYWORDS
Entrepreneurial alertness; Ghana; higher education; learner innovative behaviour; university support

Introduction

Innovation education is necessary for preparing higher education learners to be successful entrepreneurs in the future. This assertion is made given that innovation education equips learners in higher education with the requisite knowledge and skills to identify viable business opportunities and effectively allocate resources for launching and sustaining business ventures over the long term (Kruger & Steyn, 2024; Lee & Yuan, 2018). Rooted in the Iceland Design and Craft tradition, the innovation education paradigm underscores the need to address environmental concerns and challenges and potentially enhance or redesign existing products or solutions (Thorsteinsson & Denton, 2003). Our contextual definition of innovation

CONTACT Faisal Iddris educationfai2017@gmail.com Department of Management Studies Education, Akenten Appiah-Menka University of Skills Training and Entrepreneurial Development (AAMUSTED), P.O Box 1277, Kumasi, Ghana

education in this study is consistent with Thorsteinsson's (2013) assertion. Thorsteinsson defines innovation education as a structured approach to nurturing ideation skills, encompassing activities such as identifying needs, brainstorming, generating and selecting initial solutions, concept drawing or modelling to develop solutions, creating a description of the solution, and presenting ideas with the overarching goal of enhancing innovativeness of students. The innovation education tradition transcends disciplinary boundaries, emphasising experiential learning, collaboration, and application of design thinking to real-world challenges (Chandra et al., 2021; Thorsteinsson, 2013). We use innovation education in the present study as a generic concept encompassing the training and development of higher education learners that nurture their innovative thinking and ideas to contribute meaningfully to national development (Córdoba-Pachón et al., 2021). We thus argue that through innovation education, higher education learners should actively engage in addressing real-world issues, conceptualising solutions, and researching to develop the necessary knowledge for entrepreneurship (Rosa & Mujiarto, 2020; Thorsteinsson, 2013). The innovation education approach fosters a learner mindset that encourages the utilisation of various tools to pursue novel ventures (Iddris et al., 2023).

The need to use innovation education to leverage learner innovative behaviour in higher education has become imperative due to the ever-increasing complex nature of today's world (Alshuhumi et al., 2024; Biney & Salifu, 2023; N. N. Nguyen et al., 2024; Salifu & Owusu-Acheampong, 2021). As used in this study, Learner Innovative Behaviour refers to the actions and attitudes exhibited by students in environments where universities cultivate an innovation-oriented culture (Scott & Bruce, 1994). In the context of this study, it encompasses students' capabilities and willingness to engage in innovative endeavours. Moreover, according to Gilbert et al. (2021), the rapid evolution of knowledge paradigms, the shifting role of educational technologies, and the evolving needs of students all emphasise the importance of innovation in higher education. With rapid technological advancements, globalisation, and socio-economic challenges, capacity building of learners in higher education for innovative thinking and adaptability is vital for developmental goals and global competitiveness, especially in the developing world (Akyeampong, 2014).

In Ghana, recent trends in higher education suggests the sector is rapidly developing its potential to transform academic culture towards creativity, problem-solving, and entrepreneurship (Iddris et al., 2023; Kirzner, 1973). Integrating innovation education into the country's higher education curriculum is timely to help prepare learners to be change agents capable of identifying opportunities and driving sustainable development initiatives in the country. While innovation education holds significant potential for driving economic development in developing countries like Ghana, its specific impact on learner innovative behaviour within Ghanaian higher education remains largely unexplored due to a lack of research in this area. A burgeoning body of existing literature (e.g. Ahmad et al., 2023; Huang et al., 2023; Iddris et al., 2023; Johnson & Brown, 2018; Kruger & Steyn, 2024; Romani-Torres & Norena-Chavez, 2023; Shen et al., 2021; Wibowo et al., 2023) acknowledges the broader economic benefits of innovation education, yet the direct correlation with learner innovative behaviour context remains unclear. Without empirical evidence specific to Ghana, policymakers and educators lack a clear understanding of the effectiveness of innovation education initiatives in fostering learner innovative behaviour among university students.

Moreover, the limited research leaves unresolved questions regarding the mediating role of entrepreneurial alertness and the moderating influence of university support in the relationship between innovation education and learner innovative behaviour. This uncertainty hampers both scholarly understanding and practical applications of innovation education initiatives in Ghanaian higher education. To address these gaps, our study intends to investigate the effect of innovation education on learner innovative behaviour within Ghanaian higher education. It also seeks to assess the mediating effect of entrepreneurial alertness and the moderating effect of university support on the innovation education – learner innovative behaviour relationship. Through this research, we aim to make contributions to the existing literature by highlighting how fostering entrepreneurial alertness and implementing university support structures align with the primary objectives of many higher education institutions committed to fostering entrepreneurship and innovation among their student populations. These strategies can be integrated into the existing frameworks of universities across the world, especially those in the developing countries, to enhance the impact of their innovation education initiatives.

Based on the study objectives, we posed the following questions:

RQ1. What is the effect of innovation education on learner innovative behaviour in Ghanaian higher education?

RQ2. What mediation role does entrepreneurial alertness play between innovation education in Ghanaian higher education and learner innovative behaviour?

RQ3. What moderation role does university support play in the relationship between learner innovative in Ghanaian higher education and learner innovative behaviour?

Literature review

Theoretical underpinning

The present study integrates entrepreneurial alertness (Kirzner's, 1973) and the Resource-Based View (Barney, 1991; Penrose, 1959) theories to provide a comprehensive perspective on the effect of innovation education on learner innovative behaviour in Ghanaian higher education. While entrepreneurial alertness theory posits that entrepreneurial success depends on entrepreneurs' ability to spot and exploit opportunities in the market, the resource-based view emphasises identifying and integrating resources to create an innovative environment within higher education institutions. University support, such as access to mentors, research facilities, and funding, are valuable resources that enhance an individual's capacity to translate innovation education into innovative behaviour. Overall, the preoccupation in the present study was to examine the interplay between innovation education and learner innovative behaviour with entrepreneurial alertness as a mediator and University support as a moderator in the context of Ghanaian higher education. By unravelling these dynamics, the study aims to inform policy and practice interventions for nurturing a culture of innovation and entrepreneurship vital for Ghana's socio-economic advancement.

Hypothesis development

In this section, we present the study's hypotheses depicted in Figure 1 titled Conceptual Framework for understanding the relationship between innovation education and learner innovative behaviour.

How does innovation education relate to learner innovative behaviour?

Midgley and Dowling (1978) explored innovativeness, understanding it as proactive behaviours involving seeking solutions, problem-solving enthusiasm, and embracing change. Education, as Dunne (2021) outlines, shapes cognition, emotions, and behaviour through skills acquisition and idea generation. Equipping higher education learners with a business-oriented and innovative mindset nurtures entrepreneurial aspirations (Borgen & Dalgaard, 2021). Innovation Education, instructing learners in cultivating innovative skills, plays a vital role (Córdoba-Pachón et al., 2021).

University learners, with comprehensive training, possess potential for ventures, further honed through innovation education (Saji & Nair, 2018). Iddris et al. (2022) highlight a direct relationship between innovation education and entrepreneurial intent. Laguía et al. (2019) noted that postgraduates' intention for businesses linked to innovation education. Innovation education reduces the daunting aspects of starting new enterprises, fuelling entrepreneurial ambitions (Laguía et al., 2019; Souitaris et al., 2007; Wilson et al., 2007).

Innovation education positively influences learner innovative behaviour, supported by research (Kuratko et al., 2015; Shen et al., 2021). The RBV emphasises innovation education as a valuable resource for harnessing capabilities for competitive advantage. Investing in innovation education enhances learners' innovative behaviour by providing valuable knowledge and skills for generating new ideas, products, or processes. Therefore, we hypothesis that:

H1: Innovative Education has positive effect on Learner Innovative Behaviour.

The mediating role of entrepreneurial alertness

Innovation education plays a pivotal role in providing essential knowledge and skills for fostering innovative practices (Dabbagh, 2005). Entrepreneurial alertness, deeply rooted in entrepreneurial literature (Kirzner, 1973), enables individuals to discern and capitalise on innovation opportunities (Casson, 1982). While entrepreneurial alertness empowers individuals to identify opportunities and take decisive actions (Kirzner, 1973), innovation education equips learners to conceive, develop, and execute innovative ideas (Dabbagh, 2005).

Research by Alvi et al. (2017) suggests that entrepreneurial alertness directly and indirectly influences entrepreneurial intent. Moreover, Jiao et al. (2014) found that

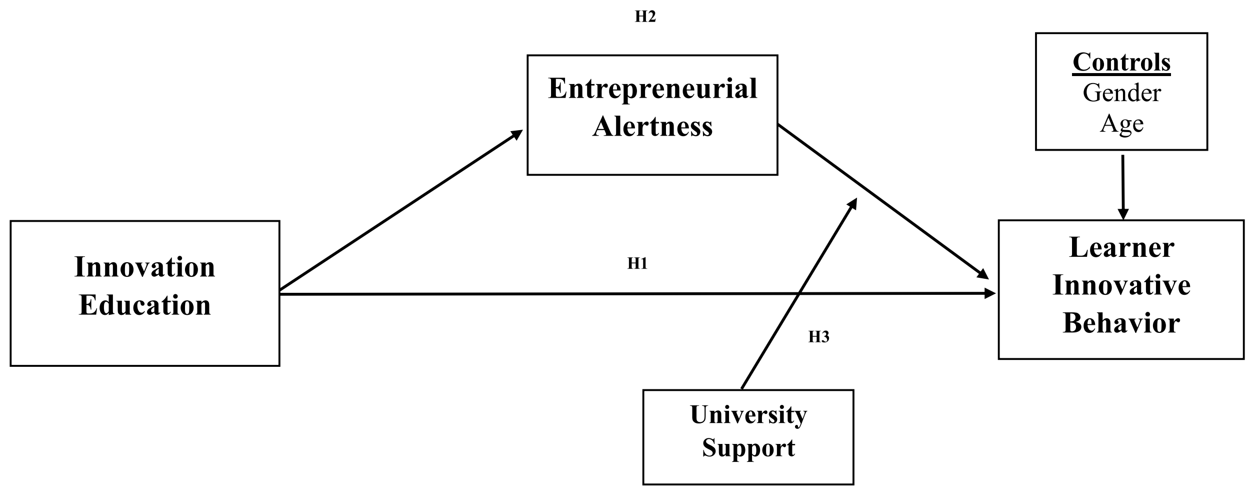

Figure 1. Conceptual framework on innovation education and learner innovative behaviour.

entrepreneurial alertness acts as a mediator between knowledge acquisition and entrepreneurs' innovativeness. Individuals undergoing innovation education are thus inclined to exhibit high entrepreneurial alertness, fostering enhanced innovation behaviour.

Kirzner's (1973) defines entrepreneurial alertness as the ability to identify and seize entrepreneurial opportunities. This implies that learners' level of entrepreneurial alertness mediates the relationship between exposure to innovation education and subsequent demonstration of innovative behaviour. Therefore, individuals who undergo innovation education are likely to exhibit enhanced innovative behaviour due to their increased entrepreneurial alertness. Thus, we hypothesised that:

H2: Entrepreneurial alertness is a mediator of the relationship between Innovative Education and Learner Innovative Behaviour.

The moderating role of university support

Perceived university support significantly influences innovative behaviour, reflecting individuals' perceptions of the institution's provision of resources and opportunities for entrepreneurship and innovation (Nguyen, 2023). As higher education institutions increasingly foster entrepreneurship and innovation, initiatives like entrepreneurship courses and campus-based incubators stimulate students' interest in innovation (Anjum et al., 2021), shaping a conducive environment for entrepreneurial exploration. Entrepreneurial alertness, crucial for seizing innovative opportunities (Kirzner, 1973), relies on university support for its translation into tangible innovative behaviour. Access to resources like mentorship and funding enhances this process. Drawing from the RBV, which emphasises resource utilisation for competitive advantage, university support plays a pivotal role. Thus, it moderates the relationship between innovation education and learner innovative behaviour, amplifying or dampening its effects based on resource availability. Based on the above arguments, we therefore hypothesised that:

H3: Perceived university support moderate the relationship between entrepreneurial alertness and learner innovative behaviour.

Source: Authors' own work

Figure 1 illustrates a conceptual framework designed to serve as an analytical tool for both data collection and analysis in the investigation This framework provides a visual representation of the hypothesised relationships among key variables, guiding the study's empirical exploration and interpretation.

Methods

This study used the quantitative approach and relied on a cross-sectional research design (X. Wang & Cheng, 2020). The study employed the Structural Equation Modelling (SEM) technique to conduct a cause-and-effect analysis.

We used purposive sampling technique (Creswell, 2013) to sample only students in their third year of study ($N = 574$) taking a three-hour credit course titled 'Innovation and Design Thinking (MGT365)' at a newly established university in Ghana because they were key informants. The course offers a hands-on approach to creativity, ideation, and problem-solving. Some of the course objectives include (a) applying design thinking principles and methodologies to identify opportunities for innovation and navigate through complex problem spaces and (b) engaging in the process of generating, prototyping, and rigorously testing innovative solutions through iterative design cycles, fostering adaptability and refinement. We administered questionnaires in person rather than using research assistants because we thought doing so would engender maximum cooperation. The data collection covered seven (7) days.

We followed ethical standards by applying for and obtaining ethical clearance from the institution's ethics committee. We also obtained verbal consent from the participants and the course teacher. We concealed participants' identities and assured them of the opportunity to withdraw from the study at any stage.

The questionnaire had five sections. In the first section (Section A), we assessed demographic backgrounds, including age and gender. In section B, we provided five measurement items assessing Innovation Education according to Iddris et al. (2022). Section C focused on Entrepreneurial Alertness using three measurement items adapted from Saadat et al. (2022) and Li et al. (2020). In Section D, we measured university support using four measurement items suggested by Nzilano et al. (2022) and Su et al. (2021). The last section (Section D) examined Learner Innovative Behavior using four items adapted from J. Lee et al. (2019) and Y. Wang et al. (2021). Table 1 provides the profile of the participants.

Factor analysis

First, using SPSS (v.25), we performed exploratory factor analysis (EFA) to determine if the measurement items correctly loaded onto the respective unobserved variables. This study included four unobserved variables: (a) innovation education, (b) entrepreneurial alertness, (c) university support and (d) learner innovative behaviour. Measurement items that cross-loaded on distinct constructs or loaded poorly (less than 0.5) were eliminated from further analysis based on the EFA. This approach led to the removal of one measuring item from entrepreneurial alertness. The total variance extracted (TVE) from the findings of the exploratory factor analysis (EFA) was 61.013%, which was greater than the minimum predicted score of 50%. Kaiser-Meyer-Olkin (KMO) was used to test sample

Table 1. Profile of respondents.

Variable	Responses	*N*	%
Gender	Male	301	52.4
	Female	273	47.6
	Total	*574*	*100.0*
Age	Below 21	47	8.2
	21–25	367	63.9
	26–30	134	23.3
	31–35	19	3.3
	Above 35	7	1.2
	Total	*574*	***100.0***

adequacy. Its score of 0.865 was higher than the minimum predicted score of 0.6. In order to demonstrate that there was sufficient correlation between the measurement items to be eligible for EFA, Bartlett's Test of Sphericity was expected to be statistically significant. The study's findings showed a significance level of ($X2 = 366.129$; $p < 0.01$). The study's determinant of 0.002, which indicates that positive definiteness in the dataset was achieved, indicates that the correlation determinant should not equal zero (0).

We used the retained variables from the EFA in Amos (v.23) to perform Confirmatory Factor Analysis (CFA) (see Figure 2). All factor loadings depicted in Figure 2 surpass the threshold of 0.3 or 0.4, signifying a significant relationship between the observed variables and the latent constructs (IE, AE, US, and LIB). As with the EFA, this study's standardised factor loadings from the CFA were anticipated to be at least 0.5, as seen in Table 2. All measuring items strongly explained their unobserved variables; the minimal factor loading under innovation education was (0.664), that under university support was (0.665), and that under entrepreneurial alertness was (0.516) and that under learner innovative behaviour was (0.648)

With a minimum expected alpha score of 0.7, Cronbach's Alpha (CA) was computed using the retained variables using SPSS (v. 25). We accomplished this for each of the four unobserved variables, suggesting that the measuring items had a high level of internal

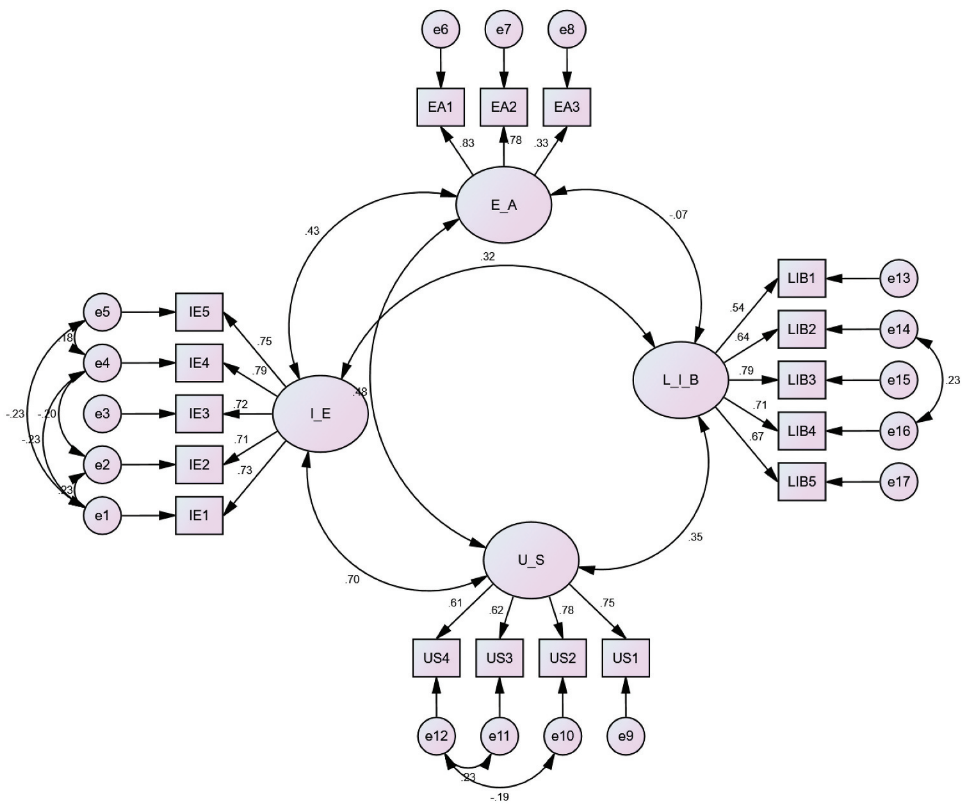

Figure 2. Diagrammatic presentation of CFA.

Table 2. Confirmatory factor analysis (CFA).

MODEL OF FITNESS: CMIN = 160.413; DF = 105; CMIN/DF = 1.528; GFI = 0.969; PClose = 1.000; TLI = 0.980; CFI = 0.984; RMSEA = 0.030; RMR = 0.048	Standardized Factor Loadings
INNOVATION EDUCATION (IE): CA = .851; CR = .859; AVE = .550	
I have had classroom education on the concept of innovation (IE1)	.664
I have been taught on the need to be innovative in business operations (IE2)	.711
My knowledge on innovation has been assessed through formal examination (IE3)	.773
I have had classroom education on the concept of creativity (IE4)	.766
As part of my course assessment, I have ever been tasked to development a new product/ service for an commercialisation (IE5)	.786
UNIVERSITY SUPPORT (US): CA = .788; CR = .808; AVE = .515	
My university offers internship focusing on entrepreneurship and innovation (US1)	.687
My university offers a bachelor's or master's course in entrepreneurship (US2)	.665
My university arranges seminars and workshops on entrepreneurship and innovation (US3)	.799
My university brings entrepreneurial students into contact with each other (US4)	.711
ENTREPRENEURIAL ALERTNESS (EA): CA = .750; CR = .777; AVE = .547	
I have frequent interactions with others to acquire new information (EA1)	.826
I always keep an eye out for new business ideas when looking for information (EA2)	.832
I have a gut feeling for potential opportunities (EA3)	.516
I can distinguish between profitable opportunities and not-so-profitable opportunities.	D*
LEARNER INNOVATIVE BEHAVIOUR (LIB): CA = .809; CR = .859; AVE = .551	
I develop ideas in new ways to solve difficult problems (LIB1)	.648
I try to find new ways to be used to perform my work (LIB2)	.775
I consider creative solutions to work-related problems (LIB3)	.788
I will introduce my new ideas to people around me to gain support and recognition (LIB4)	.784
I often come up with some innovative ideas (LIB5)	.707

D* - Items deleted due to poor factor loadings.

Table 3. Discriminant validity.

Variables	Gender	Age	Program	IE	EA	US	LIB
Gender	-						
Age	−.027	-					
Program	−.003	−.021	-				
IE	−.017	−.008	−.004	*.742*			
EA	−.014	.030	.002	.320**	*.738*		
US	−.060	−.063	.001	.446**	.361**	*.718*	
LIB	.052	−.016	−.029	.125**	−.033	.143**	*.742*

*** ~ Correlation is significant at the 0.01 level (2-tailed); * ~ Correlation is significant at the 0.05 level (2-tailed); √AVE ~ Bold and Italics.*

consistency or reliability (see Table 2). In order to ascertain if the dataset accurately fits the estimated model, it was crucial to evaluate the model fit indices during the CFA process. According to Hair et al. (2010), the model's fitness should include the following criteria: TLI and CFI should be more than 0.9, PClose should not be statistically significant (> than 0.05), CMIN/DF should be less than 3, and RMSEA and RMR should also be less than 0.08. For every unobserved variable in this dataset, these were attained (see Table 2).

The discriminant validity (see Table 3) was obtained by comparing the square root of raw average variance extracted (√AVE) to their corresponding correlation coefficients. √AVE is shown in bold and italics, which were all larger than the respective correlation coefficients. The least √AVE was 0.718 and the highest correlation was 0.446 (innovation education and university support). Results as presented in Table 3 also suggest that the highest correlation score in the entire model was 0.446 (innovation education and university support), which was less than 0.7, and was concluded there was no multicollinearity in the dataset. In conclusion, data obtained from CFA analysis is legitimate for structural model estimation.

Results

We estimated the path coefficients using covariance-based Structural Equation Modelling (SEM) in Amos (v.23) software. We used Bias-Corrected (BC) percentile method of bootstrapping, with 5000 bootstrap samples, and a 95% confidence level. Table 4 displays the outcomes of the analysis, while Figure 3 illustrates and explains the direct and mediation path estimates within the Structural Equation Modeling (SEM) framework. It offers an interpretation of the relationships among variables in the model, shedding light on the mechanisms underlying the observed associations.

Table 4. Direct and mediation path estimates.

Direct Paths	Unstandardized Estimate	S.E.	C.R.
I_E → E_A	.511	.511	7.724***
I_E → L_I_B	.181	.181	4.733***
U_S → L_I_B	.214	.214	6.473***
E_A → L_I_B	−.147	−.147	−4.462***
Gender → L_I_B	.059	.059	1.389
Age → L_I_B	−.007	−.007	−.244
Interaction_USxEA → L_I_B	.095	.095	5.016***
Indirect Effect		*Lower BC*	*Upper BC*
I_E → E_A → L_I_B	−.075	−.119	−.039

Bootstrap Bia-Corrected Confidence Interval at 95%.
***Sig. at 1%.

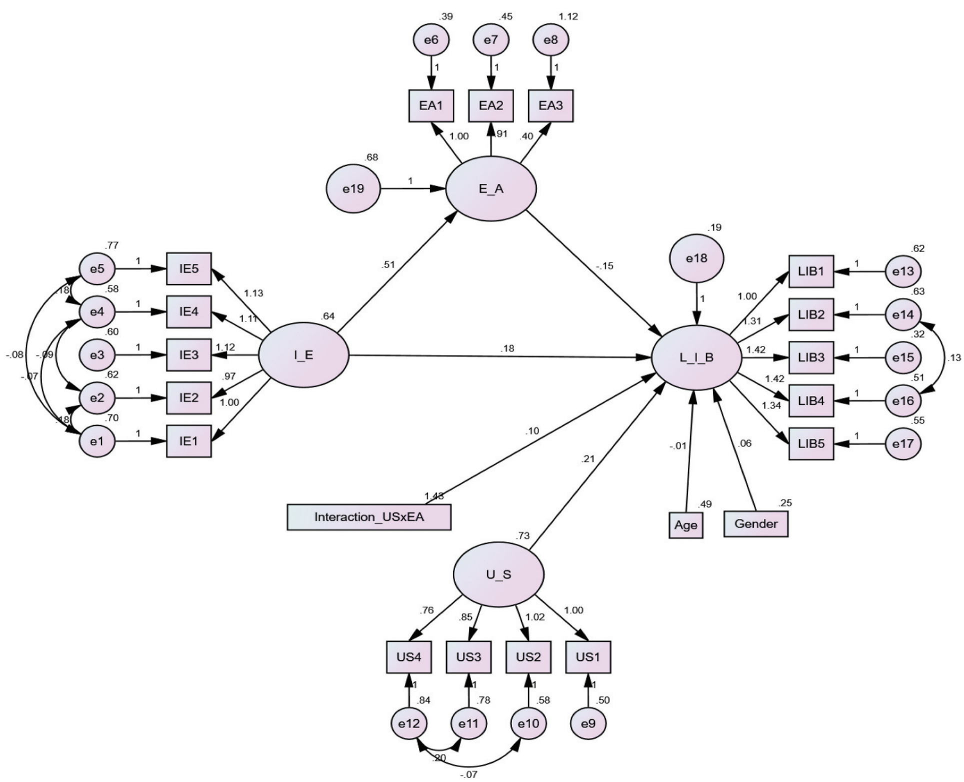

Figure 3. Structural equation model.

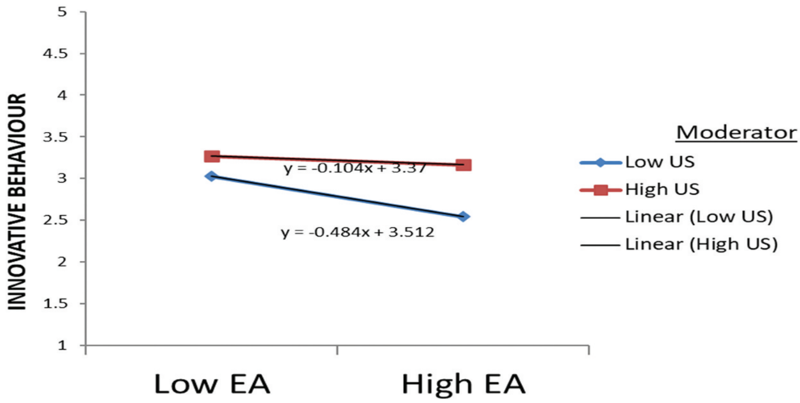

Figure 4. Interaction between entrepreneurial alertness and learner innovative behaviour.

The study controlled for gender and age. From the results presented, age had a positive but statistically insignificant effect on learner innovative behaviour (β=-.007;$p < 0.05$). Gender had significant positive effect on learner innovative behaviour (β = 0.059; $p < 0.05$). Gender was coded as 1-male and 2-female.

Regarding the main path analysis, *H1* posited that innovation education positively influences learner innovative behaviour. The study revealed a significant positive effect of innovation education on learner innovative behaviour among university students (β = 0.181; $p < 0.01$), indicating that effective innovation education enhances learner innovative behaviour by approximately 18.1%. In addition, *H2*, suggesting that entrepreneurial alertness mediates the relationship between innovation education and learner innovative behaviour, was confirmed. The indirect effect of innovation education on learner innovative behaviour through entrepreneurial alertness was found to be significantly positive (β = 0.511; $p < 0.01$), indicating that effective innovation education improves learners' entrepreneurial alertness by around 51.1%. However, the effect of entrepreneurial alertness on learner innovative behaviour was significantly negative (β=-0.147; $p < 0.01$), suggesting that entrepreneurial alertness tends to reduce learner innovative behaviour by about 14.7%. Consequently, entrepreneurial alertness was identified as a partial mediator in the relationship between innovation education and learner innovative behaviour, as both lower BC (−0.119) and upper BC (−0.039) were negative (see Table 4), supporting *H2*. Furthermore, as hypothesised in *H3*, the moderating effect of university support on the relationship between innovation education and learner innovative behaviour was examined. The interaction between entrepreneurial alertness and university support demonstrated a significant positive effect on learner innovative behaviour (β = 0.095; $p < 0.01$). Both high levels of entrepreneurial alertness and university support resulted in improved learner innovative behaviour, while low levels of both factors led to a decline (see Figure 4). Thus, the study supports hypothesis *H3*, which states that university support positively moderates the relationship between innovation education and learner innovative behaviour.

Discussion

This study contributes to the understanding of the complex relationship between innovation education and learner innovative behaviour. It also highlights how entrepreneurial alertness mediates this relationship and how support systems in the higher education sector moderate it. By drawing on theoretical frameworks of entrepreneurial alertness and resource-based view, this research sheds light on the interplay of these factors and their collective influence on learner innovative behaviour. The findings mainly affirm the study's hypotheses and support the prior research conducted by Iddris et al. (2022) and Kuratko et al. (2015)

The initial hypothesis *(H1)* posited that innovation education influenced learner innovative behaviour positively. Through path analysis, results have validated this hypothesis by indicating a significant positive effect of innovation education on learner innovative behaviour. These results corroborate those of previous research, such as Kim et al. (2018), Kuratko et al. (2015) and Lv et al. (2022), suggesting that exposure to innovation education fosters advancements in learner innovative behaviour. The assertion emphasises the potential of innovation education to cultivate students' innovative capabilities. For instance, Kim et al. (2018) conducted a study examining the effectiveness of incorporating design thinking principles into engineering education. They found that students exposed to innovative teaching methods demonstrated higher levels of creativity and problem-solving skills, indicating the beneficial effects of innovation education on learner innovative behaviour.

Moreover, the study confirmed *H2*, suggesting entrepreneurial alertness mediated the relationship between innovation education and learner innovative behaviour. This finding is consistent with existing literature emphasising the pivotal role of alertness in identifying business opportunities (Lew et al., 2023; Saadat et al., 2022; Wibowo et al., 2023). However, the revelation of the present study that entrepreneurial alertness did not directly contribute to learner innovative behaviour negates the outcome of Araujo et al. (2023) study, which identified innovation as a crucial outcome of alertness. Based on their findings, the researchers concluded that support services should enhance alertness which in turn increases learner innovative behaviour. Again, our finding also contradicts the findings of Kadile and Biraglia (2020) that environmental factors such as feedback, collaboration offers, and awards positively impact on alertness. In effect, these two groups of researchers appear to suggest that innovation education enhances students' alertness, thereby contributing to their journey towards creativity and innovation.

Lastly, *H3* was supported, indicating that while innovation education positively impacted learner innovative behaviour, this relationship is strengthened by perceived university support. This finding reinforces the views of Anjum et al. (2021) and Sampene et al. (2023), which emphasis the moderating effect of perceived university support in the relationship between entrepreneurship education and entrepreneurial intention. Anjum et al. (2021) conducted a study examining the influence of university support services on students' entrepreneurial intentions. They found that students who perceived greater support from their institutions were more likely to engage in entrepreneurial activities, highlighting the importance of supportive environments in fostering innovation.

Thus far, the results demonstrate that while innovation education is a fundamental predictor of learner innovative behaviour, relying solely on it falls short. Empirical evidence from the present study shows a significant negative relationship between entrepreneurial alertness and learner innovative behaviour. However, the moderation of university support is crucial for creating a positive relationship between entrepreneurial alertness and learner innovative behaviour. Thus, as Kirzner's (1973) alertness theory emphasises, entrepreneurial alertness may not necessarily translate into tangible learner innovative behaviour unless there is a deliberate effort to establish university support systems to support learners who exhibit entrepreneurial alertness. Combining university support and entrepreneurial alertness could lead to learner innovativeness (Brandão Farias et al., 2024). The results support the proposition that high levels of entrepreneurial alertness facilitate the translation of knowledge gained through innovation education into tangible innovative outcomes (Shane & Venkataraman, 2000). The results also extend the resource-based view (Barney, 1991) into the realm of innovation education by highlighting the role of university support as a valuable resource (Guerrero et al., 2016) that moderates the innovation process. Again, the results demonstrate the importance of considering external factors in the innovation ecosystem that interact with internal cognitive processes, thus enriching our theoretical understanding of learner innovative behaviour.

Conclusion and implications

The present study examined the effect of innovation education on learner innovative behaviour. It also assessed the mediating role of entrepreneurial alertness and the moderating role of university support in the relationship between innovation education and learner innovative behaviour. Findings revealed innovation education as a predictor of learner innovative behaviour in higher education. The findings further demonstrated that although a connection existed between innovation education and learner innovative behaviour, entrepreneurial alertness and university support were critical assets for fostering their relationship.

The findings offer several implications. First, the study suggests that innovation education alone may not suffice to influence learner innovative behaviour. Therefore, integrating entrepreneurial alertness and university support alongside innovation education is crucial to enhancing their relationship. This insight expands existing literature and provides valuable insights into the complex dynamics between innovation education and learner innovative behaviour. Globally, the findings highlight the important role of Higher Education Institutions in nurturing entrepreneurially alert learners and fostering an environment conducive to learner innovative behaviour. The findings suggest that innovation education has the potential to drive economic development in developing country contexts. By investing in innovative education initiatives and supporting student innovators, universities can contribute to the growth of local economies and address broader societal challenges.

Second, the positive relationship between innovation education and learner innovative behaviour highlights the importance of prioritising the integration of innovation-focused curricula and pedagogical approaches into educational programmes offered by higher education institutions. By incorporating workshops, entrepreneurship and innovation courses, and resources conducive to fostering a culture of continuous learning and innovation, higher education institutions can effectively enhance learner innovative behaviour and equip students with the necessary knowledge and skills for innovation. This integration should extend to employee training programmes, enabling managers to foster a culture of innovation within the workplace.

Third, higher education institutions should recognise the significance of entrepreneurial alertness in fostering innovation among students. Developing training programmes and workshops aimed at enhancing students' alertness to entrepreneurial opportunities can empower them to identify and capitalise on innovative ideas within their academic and professional pursuits. Raising awareness and providing practical exercises can further support students in recognising and seizing entrepreneurial opportunities.

Fourth, higher education institutions may facilitate cross-disciplinary collaboration and establish innovation hubs where students from diverse academic backgrounds can collaborate, exchange ideas, and access resources and support. By fostering a culture of collaboration and innovation, universities can create an environment conducive to the development of innovative solutions to real-world challenges.

Despite the significance of this study, it has some limitations. Firstly, the use of a sample from one purposively selected Ghanaian university may limit the generalisability of the findings. Future research should consider expanding the scope to include higher education institutions across the country for enhanced generalisability. Additionally, the cross-sectional design of the study limits the establishment of causality. Future longitudinal studies can track the evolution of the influence of innovation education, entrepreneurial alertness, and university support on learner innovative behaviour, establishing causality and providing insights into long-term effects.

Furthermore, future research could explore other mediating factors influencing the relationship between innovation education, university support, and learner innovative behaviour, such as innovation mindset, resilience, self-efficacy, innovation culture, risk-taking propensity, technological proficiency, and emotional intelligence. Lastly, despite not finding a significant effect of entrepreneurial alertness on innovative behaviour, further investigation is warranted to understand the conditions under which entrepreneurial alertness is most effective in promoting learner innovative behaviour.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Faisal Iddris PhD, is an Associate Professor of Innovation Management and administrator currently serving as the Dean of the Faculty of Business Education at AAMUSTED. Dr. Iddris has published over 30 scientific articles and conference articles

Inusah Salifu is an Associate Professor at the Department of Adult Education and Human Resource Studies, University of Ghana. His research interests cover teachers and teaching, learners and learning, and general educational administration/management and leadership at the tertiary and pre-tertiary levels.

Emmanuel Mensah Kparl is a researcher and academic affiliated with the Department of Management Studies Education at AAMUSTED. He currently serves as a teaching and research assistant in the department. Mr. Kparl has co-authored several scholarly articles in the fields of business, innovation, entrepreneurship education, and management.

ORCID

Inusah Salifu http://orcid.org/0000-0002-9626-6211
Emmanuel Mensah Kparl http://orcid.org/0000-0002-8260-4822

References

Ahmad, N., Youjin, L., Žiković, S., & Belyaeva, Z. (2023). The effects of technological innovation on sustainable development and environmental degradation: Evidence from China. *Technology in Society*, *72*, 102184. https://doi.org/10.1016/j.techsoc.2022.102184

Akyeampong, K. (2014). Reconceptualised life skills in secondary education in the African context: Lessons learnt from reforms in Ghana. *International Review of Education*, *60*(2), 217–234. https://doi.org/10.1007/s11159-014-9408-2

Alshuhumi, S. R., Al-Hidabi, D. A., & Al-Refaei, A. A. A. (2024). Unveiling the behavioural nexus of innovative organizational culture: Identification and affective commitment of teachers in primary schools. *Journal of Human Behaviour in the Social Environment*, *34*(1), 130–152. https://doi.org/10.1080/10911359.2023.2267600

Alvi, I., Sharma, A., & Alvi, T. (2017). Influence of entrepreneurial alertness of professional students on entrepreneurial intentions and determinants of entrepreneurial intentions. *Amity Journal of Entrepreneurship*, *2*(1), 32–46.

Anjum, T., Heidler, P., Amoozegar, A., & Anees, R. T. (2021). The impact of entrepreneurial passion on the entrepreneurial intention; moderating impact of perception of university support. *Administrative Sciences*, *11*(2), 45. https://doi.org/10.3390/admsci11020045

Araujo, C. F., Karami, M., Tang, J., Roldan, L. B., & dos Santos, J. A. (2023). Entrepreneurial alertness: A meta-analysis and empirical review. *Journal of Business Venturing Insights*, *19*, e00394. https://doi.org/10.1016/j.jbvi.2023.e00394

Barney, J. B. (1991). The resource based view of strategy: Origins, implications, and prospects. *Journal of Management*, *17*(1), 97–98. https://doi.org/10.1177/014920639101700107

Biney, I. K., & Salifu, I. (2023). Opportunities for adult learners in Ghana's higher education institutions: Limiting factors and strategies. *Journal of Continuing Higher Education*, *72*(2), 157–171. https://doi.org/10.1080/07377363.2023.2178664

Borgen, C., & Dalgaard, B. R. (2021). Business education in the legacy of Hans Nielsen Hauge. In *The legacy of Hans Nielsen Hauge. Puritan economics, entrepreneurship, and management* (pp. 216). Bodoni forlag.

Brandão Farias, A. C., Moraes, G. H. S. M. D., de Campos, M. L., Cappellozza, A., Fischer, B. B., & Anholon, R. (2024). The moderating effect of the university perceived support environment on entrepreneurial behaviour. *Innovations in Education and Teaching International*, *61*(1), 85–98. https://doi.org/10.1080/14703297.2022.2152361

Casson, M. (1982). *The entrepreneur: An economic theory*. Rowman & Littlefield.

Chandra, P., Tomitsch, M., & Large, M. (2021). Innovation education programs: A review of definitions, pedagogy, frameworks and evaluation measures. *European Journal of Innovation Management*, *24*(4), 1268–1291. https://doi.org/10.1108/EJIM-02-2020-0043

Córdoba-Pachón, J. R., Mapelli, F., Taji, F. N. A. A., & Donovan, D. M. (2021). Systemic creativities in sustainability and social innovation education. *Systemic Practice and Action Research*, *34*(3), 251–267. https://doi.org/10.1007/s11213-020-09530-z

Creswell, J. W. (2013). *Qualitative, quantitative, and mixed methods approach* (4th ed.). Sage.

Dabbagh, N. (2005). Pedagogical models for e-learning: A theory-based design framework. *International Journal of Technology in Teaching and Learning*, *1*(1), 25–44.
Dunne, J. (2021). What's the good of education? In W. Carr (Ed.), *The Routledge Falmer reader in philosophy of education* (pp. 145–160). Routledge.
Gilbert, A., Tait McCutcheon, S., & Knewstubb, B. (2021). Innovative teaching in higher education: Teachers' perceptions of support and constraint. *Innovations in Education and Teaching International*, *58*(2), 123–134. https://doi.org/10.1080/14703297.2020.1715816
Guerrero, M., Urbano, D., & Fayolle, A. (2016). Entrepreneurial activity and regional competitiveness: Evidence from European entrepreneurial universities. *The Journal of Technology Transfer*, *41*(1), 105–131. https://doi.org/10.1007/s10961-014-9377-4
Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2010). *Multivariate data analysis: A global perspective* (7th ed.). Pearson Education, Inc.
Huang, L., Bai, X., Huang, L., Huang, Y., & Han, G. (2023). How does college students' entrepreneurial learning influence entrepreneurial intention: Evidence from China. *Sustainability*, *15*(12), 9301. https://doi.org/10.3390/su15129301
Iddris, F., Dogbe, C. S. K., & Kparl, E. M. (2022). Innovation education and entrepreneurial intentions among postgraduate students: The role of innovation competence and gender. *Cogent Education*, *9*(1), 2083470. https://doi.org/10.1080/2331186X.2022.2083470
Iddris, F., Mensah, P. O., Adjanor-Doku, C., & Yaa Akyiaa Ellis, F. (2023). Enhancing firm innovativeness through HRM practices: The mediating role of innovation capability. *International Journal of Innovation Science*. https://doi.org/10.1108/IJIS-06-2023-0131
Jiao, H., Cui, Y., Zhu, Y., & Chen, J. (2014). Building entrepreneurs' innovativeness through knowledge management: The mediating effect of entrepreneurial alertness, technology. *Analysis & Strategic Management*, *26*(5), 501–516. https://doi.org/10.1080/09537325.2013.872774
Johnson, R. L., & Brown, S. (2018). The effects of innovation education on innovative behaviour in organizations. *Innovation Management Journal*, *21*(2), 156–172.
Kadile, V., & Biraglia, A. (2020). From hobby to business: Exploring environmental antecedents of entrepreneurial alertness using fsQCA. *Journal of Small Business Management*, *60*(3), 580–615. https://doi.org/10.1080/00472778.2020.1719846
Kim, J. Y., Choi, D. S., Sung, C. S., & Park, J. Y. (2018). The role of problem solving ability on innovative behaviour and opportunity recognition in university students. *Journal of Open Innovation: Technology, Market, and Complexity*, *4*(2), 4. https://doi.org/10.3390/joitmc4020004
Kirzner, I. M. (1973). *Competition and entrepreneurship*. University of Chicago Press.
Kruger, S., & Steyn, A. A. (2024). Developing breakthrough innovation capabilities in university ecosystems: A case study from South Africa. *Technological Forecasting & Social Change*, *198*, 123002. https://doi.org/10.1016/j.techfore.2023.123002
Kuratko, D. F., Morris, M. H., & Schindehutte, M. (2015). Understanding the dynamics of entrepreneurship through framework approaches. *Small Business Economics*, *45*(1), 1–13. https://doi.org/10.1007/s11187-015-9627-3
Laguía, A., Moriano, J. A., & Gorgievski, M. J. (2019). A psychosocial study of self-perceived creativity and entrepreneurial intentions in a sample of university students. *Thinking Skills and Creativity*, *31*, 44–57. https://doi.org/10.1016/j.tsc.2018.11.004
Lee, J., Kim, D., & Sung, S. (2019). The effect of entrepreneurship on start-up open innovation: Innovative behaviour of university students. *Journal of Open Innovation: Technology, Market, and Complexity*, *5*(4), 103. https://doi.org/10.3390/joitmc5040103
Lee, R. M., & Yuan, Y. (2018). Innovation education in China: Preparing attitudes, approaches, and intellectual environments for life in the automation economy. In N. W. Gleason (Ed.), *Higher education in the era of the fourth industrial revolution* (pp. 93–119). Palgrave Macmillan. https://doi.org/10.1007/978-981-13-0194-0_5
Lew, Y. K., Zahoor, N., Donbesuur, F., & Khan, H. (2023). Entrepreneurial alertness and business model innovation in dynamic markets: International performance implications for SMEs. *R&D Management*, *53*(2), 224–243. https://doi.org/10.1111/radm.12558
Li, C., Murad, M., Shahzad, F., Khan, M. A. S., Ashraf, S. F., & Dogbe, C. S. K. (2020). Entrepreneurial passion to entrepreneurial behaviour: Role of entrepreneurial alertness, entrepreneurial

self-efficacy and proactive personality. *Frontiers in Psychology*, *11*, 1611. https://doi.org/10.3389/fpsyg.2020.01611

Lv, M., Zhang, H., Georgescu, P., Li, T., & Zhang, B. (2022). Improving education for innovation and entrepreneurship in Chinese technical universities: A quest for building a sustainable framework. *Sustainability*, *14*(2), 595.

Midgley, D. F., & Dowling, G. R. (1978). Innovativeness: The concept of innovation and diffusion of innovation. *The Journal of Consumer Research*, *4*(4), 229–242. https://doi.org/10.1086/208701

Nguyen, N. N., Le, T. T., Thi Nguyen, B. P., & Nguyen, A. (2024). Examining effects of students' learner innovative behaviour and problem-solving skills on crisis management self-efficacy: Policy implications for higher education. *Policy Futures in Education*, *22*(1), 1–20. https://doi.org/10.1177/14782103221133892

Nguyen, T. P. L. (2023). Factors affecting innovative behaviour of Vietnamese enterprises employees. *International Journal of Innovation Science*, *15*(1), 186–203. https://doi.org/10.1108/IJIS-09-2021-0166

Nzilano, K. L., Tundui, H. P., & Ndyetabula, D. W. (2022). Perceived university support and technical graduates' intentions to venture into business start-ups in Tanzania: Does institutional support matter? *Journal of Global Entrepreneurship Research*, *12*(1), 465–477. https://doi.org/10.1007/s40497-022-00334-0

Penrose, E. (1959). A resource-based view of the firm. *Strategic Management Journal*, *5*(2), 171–180. https://doi.org/10.1002/smj.4250050207

Romani-Torres, R., & Norena-Chavez, D. (2023). Fueling recognition of opportunities through innovative behaviour: Mediating role of team entrepreneurial passion and team innovation capacity. *Cogent Business & Management*, *10*(3), 2259580. https://doi.org/10.1080/23311975.2023.2259580

Rosa, A. T. R., & Mujiarto, M. (2020). Teacher development potential (creativity and innovation) education management in engineering training, coaching and writing works through scientific knowledge intensive knowledge based on web research in the industrial revolution and society. *International Journal of Higher Education*, *9*(4), 161–168. https://doi.org/10.5430/ijhe.v9n4p161

Saadat, S., Aliakbari, A., Alizadeh Majd, A., & Bell, R. (2022). The effect of entrepreneurship education on graduate students' entrepreneurial alertness and the mediating role of entrepreneurial mindset. *Education + Training*, *64*(7), 892–909. https://doi.org/10.1108/ET-06-2021-0231

Saji, B. S., & Nair, A. R. (2018). Effectiveness of innovation and entrepreneurship education in UAE higher education. *Academy of Strategic Management Journal*, *17*(4), 1–12.

Salifu, I., & Owusu-Acheampong, E. (2021). Improving higher education instructional delivery in the developing world: The role of university teachers as digital leaders. In M. Mohiuddin (Ed.), *LeaRdership in a changing world: A multidimensional perspective*, (Chapter 11, pp. 1–14). IntechOpen. https://doi.org/10.5772/intechopen.100546

Sampene, A. K., Li, C., Khan, A., Agyeman, F. O., & Opoku, R. K. (2023). Yes! I want to be an entrepreneur: A study on university students' entrepreneurship intentions through the theory of planned behaviour. *Current Psychology*, *42*(25), 21578–21596. https://doi.org/10.1007/s12144-022-03161-4

Scott, S. G., & Bruce, R. A. (1994). Determinants of innovative behavior: A path model of individual innovation in the workplace. *Academy of Management Journal*, *37*(3), 580–607. https://doi.org/10.2307/256701

Shane, S., & Venkataraman, S. (2000). The promise of entrepreneurship as a field of research. *Academy of Management Review*, *25*(1), 217–226. https://doi.org/10.5465/amr.2000.2791611

Shen, Y., Xie, W., Wang, X., Qu, J., Zhou, T., Li, Y., Mao, X., Hou, P., Liu, Y. (2021). Impact of innovative education on the professionalism of undergraduate nursing students in China. *Nurse Education Today*, *98*, 104647. https://doi.org/10.1016/j.nedt.2020.104647

Souitaris, V., Zerbinati, S., & Al-Laham, A. (2007). Do entrepreneurship programmes raise entrepreneurial intention of science and engineering students? The effect of learning, inspiration and resources. *Journal of Business Venturing*, *22*(4), 566–591. https://doi.org/10.1016/j.jbusvent.2006.05.002

Su, Y., Zhu, Z., Chen, J., Jin, Y., Wang, T., Lin, C. L., & Xu, D. (2021). Factors influencing entrepreneurial intention of university students in China: Integrating the perceived university support and theory of planned behaviour. *Sustainability*, *13*(8), 4519. https://doi.org/10.3390/su13084519

Thorsteinsson, G. (2013). Ideation training via innovation education to improve students' ethical maturation and social responsibility. *Journal on Educational Psychology*, *6*(4), 1–7. https://doi.org/10.26634/jpsy.6.4.2180

Thorsteinsson, G., & Denton, H. G. (2003). The development of innovation education in Iceland: A pathway to modern pedagogy and potential value in the UK. *Journal of Design & Technology Education*, *8*(3), 172–179.

Wang, X., & Cheng, Z. (2020). Cross-sectional studies: Strengths, weaknesses, and recommendations. *Chest*, *158*(1), S65–S71. https://doi.org/10.1016/j.chest.2020.03.012

Wang, Y., Chen, Y., & Zhu, Y. (2021). Promoting innovative behaviour in employees: The mechanism of leader psychological capital. *Frontiers in Psychology*, *11*, 598090. https://doi.org/10.3389/fpsyg.2020.598090

Wibowo, A., Narmaditya, B. S., Saptono, A., Effendi, M. S., Mukhtar, S., & Mohd Shafiai, M. H. (2023). Does digital entrepreneurship education matter for students' digital entrepreneurial intentions? The mediating role of entrepreneurial alertness. *Cogent Education*, *10*(1), 2221164. https://doi.org/10.1080/2331186X.2023.2221164

Wilson, S., Liber, O., Johnson, M., Beauvoir, P., Sharples, P., & Milligan, C. (2007). Personal learning environments: Challenging the dominant design of educational systems. *Journal of E-Learning and Knowledge Society*, *3*(2), 27–38.